Prime-support restrictions for a magic square of distinct squares

An AI-generated research report and verification index

Anonymous AI-generated research draft

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Abstract

This AI-generated research project studies the open question of whether a 3×3 magic square can have nine pairwise distinct positive integer squares. We study the prime support of the square root e of the center entry e^2. The main results recorded here exclude every primitive center root supported on exactly two rational primes, with arbitrary positive exponents, and exclude every exactly-three-block center for which all local Gaussian indices are balanced or extreme. The latter includes every squarefree center root pqr. For the first remaining exponent pattern p^2qr, an exact physical signature classification gives 134 classes; proved valuation, quotient, factorization, and congruence filters exclude 118 and leave 16 necessary classes. No surviving class is asserted to be realizable. We also describe bounded exact searches, used only as cross-checks. The unrestricted problem is not solved.

This report is a compact statement and proof index. A versioned companion archive contains the line-by-line Gaussian calculations, elliptic-curve descents, independent finite enumerators, scanners, and regression tests. The work has not been peer reviewed and makes no priority claim.

AI-generated mathematical result. AI systems generated most of the conjectures, proof strategies, derivations, code, and exposition under human direction. Uwe Schwarz contributed the problem choice, prompts, computing resources, coordination, and review; he does not claim authorship of the mathematical results. Repeated AI review and exact tests are not independent expert verification. The work is public to make errors discoverable and has not been peer reviewed.

Contents

1 Problem, conventions, and scope

A normal 3×3 magic square has equal row, column, and diagonal sums. The open problem asks whether all nine entries can be distinct positive integer squares. Since the center of every 3×3 magic square is one third of the common line sum, a square-valued center may be written e^2. Dividing the nine square roots by their common greatest common divisor reduces the problem to a *primitive* square without changing distinctness.

This report proves restrictions on the factorization of e. It does not prove existence or global nonexistence. In particular, the following remain open here:

- centers with four or more rational prime factors;
- exactly-three-block centers with genuine intermediate local indices;
- the 16 weighted signatures left by the p^2qr analysis;
- construction of a full square, or a global obstruction to all centers.

The literature still describes the unrestricted problem as open. Bremner gave elliptic-curve and intersection-of-quadrics formulations and strong near-solutions [1, 2]. More recent work studies Gaussian and residue reformulations [3, 4, 5], while Rome and Yamagishi settle the existence problem for orders at least four, not order three [6]. A bounded literature audit and discussion of recent claims is included in the companion archive.

2 Centered form and four offsets

Every 3×3 magic square with center e^2 has the form

$$\begin{pmatrix} e^2 + x & e^2 - x - y & e^2 + y \\ e^2 - x + y & e^2 & e^2 + x - y \\ e^2 - y & e^2 + x + y & e^2 - x \end{pmatrix}. \tag{1}$$

After interchanging the two centered coordinates and changing their signs, put $a > b > 0$. The four positive offsets of opposite pairs are

$$\mathcal{D} = \{a, b, a + b, a - b\}. \tag{2}$$

The entries are pairwise distinct exactly when these four offsets are nonzero and pairwise distinct. For every $d \in \mathcal{D}$ there are positive integers u_d, v_d satisfying

$$u_d^2 = e^2 + d, \qquad v_d^2 = e^2 - d, \tag{3}$$

and therefore

$$(u_d v_d)^2 + (d)^2 = e^4. \tag{4}$$

Thus each offset is the absolute imaginary coordinate of a Gaussian integer square of norm e^4. The two corner offsets a, b and two edge offsets $a + b, a - b$ also give four three-term relations. Deleting an edge gives a coefficient-111 relation; deleting a corner gives a coefficient-211 relation.

3 Gaussian blocks and local indices

Let $p^k \| e$. A primitive representation of (4) forces $p \equiv 1 \pmod 4$, so choose $p = \pi\bar{\pi}$ in $\mathbb{Z}[i]$. At the p-support, each offset has an index

$$j_{p,d} \in \{-k, -k+1, \dots, k-1, k\}. \tag{5}$$

The values $j = 0$ and $|j| = k$ will be called *balanced* and *extreme*. Values $0 < |j| < k$ are *intermediate*. Unique factorization in $\mathbb{Z}[i]$ gives the following local spectrum.

Lemma 3.1 (Local unit pattern)**.** *For every center prime p, at least three of the four offsets are p-adic units. Equivalently, at most one column has a nonextreme local valuation pattern. The possible exception is constrained by whether it is a corner or an edge and by the coefficient-111 or coefficient-211 relation obtained from the other three columns.*

The proof uses the pairwise differences among $a, b, a+b, a-b$. If two offsets had positive p-valuation, then the displayed linear relations force a third and finally contradict primitivity. Once three offsets are units, aligning their one-sided Gaussian factors produces an exact rational quotient. These quotients are the main bridge between local valuations and global sizes.

4 A block-balance theorem

Write $M = e/p^k$. The central quantitative result is the following.

Theorem 4.1 (Prime-power block balance)**.** *For every $p^k \| e$ in a primitive magic square of distinct squares,*

$$p^{2k} < 3M^2. \tag{6}$$

If $p \equiv 5 \pmod 8$, then the stronger inequality

$$p^k < M \tag{7}$$

holds. If $p^k \geq M$, there is exactly one nonunit offset; it is an edge, $p \equiv 1 \pmod 8$, and for its intermediate index j,

$$k \geq 3, \qquad \left\lfloor \frac{k}{2} \right\rfloor + 1 \leq |j| \leq k-1, \qquad p^{2(k-|j|)} < \sqrt{2}M. \tag{8}$$

Proof architecture. Align the three p-unit Gaussian squares in the retained 111 or 211 relation and remove their common one-sided factor. Unique factorization gives an identity

$$H = \bar{\Pi} n, \qquad n \in \mathbb{Z} \setminus \{0\}, \tag{9}$$

or the same identity for $H/2$ in the 211 case. All residual summands have the same modulus. Strict triangle inequalities give $|n|p^{2k} < 3M^2$ in the 111 case and the constant 2 in the 211 case. The valuation spectrum shows that the factor represented by n contains precisely the missing local content. If a block is at least as large as its complement, the corner case has incompatible p-valuation, while an edge case survives only with $(2/p) = 1$. This proves (6)–(8). The detailed unit choices, corner contents, and Legendre-symbol refinement are given in `research/prime-support/block-balance.md`.

An immediate consequence is that a center cannot be a prime power. More importantly, (6) can be applied simultaneously to every block; the resulting strict monomial inequalities are strong enough to eliminate many complete support patterns.

5 Centers supported on two primes

Theorem 5.1 (Two-block exclusion)**.** *There is no primitive magic square of distinct positive integer squares whose center root is*

$$e = p^k q^\ell \tag{10}$$

for distinct rational primes p, q and positive integers k, ℓ.

The proof is arithmetic, not a bounded search. Its main reductions are:

1. neither block can have a balanced exceptional offset;
2. the two exceptions must both be edges;
3. after ordering $P = p^k > Q = q^\ell$, the larger block is dominant, so $k \geq 3$ and $p \equiv 1 \pmod 8$;
4. the two edge relations share a forced sign parameter and a small odd quotient ν;
5. congruences and the coupled norm equations reduce to $|\nu| = 3$ and a genus-one curve.

In the final branch, rational parameters lead to

$$y^2 = t^4 - 10t^2 + 1. \tag{11}$$

A birational transformation gives

$$E: \quad v^2 = x(x-1)(x-3). \tag{12}$$

A self-contained 2-isogeny descent shows that $E(\mathbb{Q})$ has rank zero; good reduction at 5 and 7 bounds the torsion, and the visible points exhaust it. Every torsion point maps back to $t = 0$ or to a degenerate offset. This closes the last edge-edge branch and proves Theorem 5.1. The complete sign table, norm elimination, covering equations, local obstructions, and inverse maps appear in `research/prime-support/two-block-exponent-one.md`.

Remark 5.2. The arbitrary exponents in Theorem 5.1 matter. A squarefree semiprime exclusion alone does not imply the theorem.

6 Exactly three blocks in the balanced/extreme regime

Let

$$e = PQR = p^k q^\ell r^m \tag{13}$$

have exactly three prime-power blocks. Suppose every local index is balanced or extreme. After normalizing the three one-sided phases, a column is encoded by a vector in $\{0, \pm 1\}^3$ up to global sign. Distinct offsets require the four projective vectors to be nonzero and pairwise distinct. Every deleted triple must also be orientation-frustrated: a coherent 111 or 211 triple would give a rational Gaussian quotient whose one-sided prime supports cannot match.

The physical symmetry group is smaller than an arbitrary permutation of four columns: it preserves the pair of corners and the pair of edges. Quotienting by row permutations, corner swap, edge swap, row switches, and column conjugations leaves a finite list of physical signatures.

Theorem 6.1 (Three balanced/extreme blocks)**.** *No primitive center root supported on exactly three rational primes is admissible if, at every block, all four local indices are balanced or extreme.*

Corollary 6.2 (Squarefree three-prime exclusion)**.** *There is no primitive magic square whose center root is $e = pqr$ for three distinct rational primes.*

Finite and arithmetic parts. In the squarefree model, exact projective distinctness and frustration leave 29 physical classes, including one all-extreme Hadamard class. The all-extreme class reduces to two elliptic curves isomorphic to

$$Y^2 = X(X + 1)(X + 5), \tag{14}$$

whose rational points are only the degenerate torsion points. Thus 28 classes remain. They are excluded by a sequence of exact arguments: elementary trigonometric inequalities, Fermat's right-triangle descent, simultaneous square lemmas, Gaussian quotient comparisons, and three rank-zero elliptic curves. None of these 28 exclusions is a bounded search.

For arbitrary exponents, replace each prime by its full block size and each one-sided Gaussian prime by the corresponding power. Primitivity, non-axis conditions, projective distinctness, quotient integrality, and every descent survive unchanged. This transfers all 28 exclusions and proves Theorem 6.1. The classifiers and proof are in `three-block-squarefree.md`, `three_block_signatures.py`, and `three_block_squarefree.py`.

7 The first intermediate pattern: p^2qr

Theorem 6.1 forces any p^2qr survivor to have one intermediate p-index ± 1; its other p-indices are ± 2, while the q- and r-rows are balanced or extreme. Exact physical quotienting gives 134 weighted signature classes.

Proposition 7.1 (Verified p^2qr residue)**.** *Of the 134 necessary physical signatures for $e = p^2qr$, the proved filters in the companion archive exclude 118. Exactly 16 classes remain relative to those filters. No remaining class is asserted realizable.*

The reduction is intentionally staged so that every count is independently testable.

Stage	Total classes excluded
Exceptional-row monomial inequalities	58
Aligned rational quotients	85
Composite-block bounds	95
General common-factor heights	104
All admissible divisor comparisons	110
Factored axis bounds	113
Congruence-sharpened skew bounds	116
Discrete prime floors	117
Complementary-axis descent	118

The filters combine strict monomial inequalities with exact local data. For example, if an aligned retained triple has a common Gaussian factor, rational valuation symmetry turns its residual quotient into an integer. Factoring the remaining axis expression gives divisibility by one cyclotomic factor, while skew expressions produce a norm quotient with a congruence gap. The sharp uniform constants used in the final classifier are

$$S_2 = 241, \qquad S_6 = 433, \qquad S_4 = 1201, \qquad S_8 = 4201. \tag{15}$$

All cone comparisons use exact integer arithmetic, including constants; no floating-point feasibility decision is made. The 134-class enumerator and the arithmetic-filter implementation are independent programs. Their tests pin every intermediate count in the table.

The 16 survivors are a concrete frontier rather than evidence of existence. They retain enough sign and intermediate-index freedom that the current local quotients do not close. A productive next step would be to couple two or more of their exact Gaussian identities rather than add another independent height bound.

8 Bounded exact computation

The proof results above do not depend on search limits. Separate scanners were used to find counterexamples to intermediate lemmas and to cross-check enumerators. Their recorded negative results are:

Search	Exhausted range	Hits
Center-root 111 and 211 relations	$e \le 5 \cdot 10^8$	0
Rational-parameter relation scan	parameter height $H \le 5.5 \cdot 10^6$	0
Dominant-block identity scan	complementary block $M \le 10^6$	0
Two-block factor scan	$\max(P, Q) \le 10^{12}$	0

The largest rational-parameter run used four threads and enumerated 875,336 normalized values. It tested exactly 383,106,118,780 unordered pairs and completed in 8 hours, 42 minutes on the host used for the scan. The center-root and factor scans use different heights, so their zero counts are not interchangeable. This larger rational-parameter run is useful as evidence but cannot settle the Diophantine question.

An independent B6 project reports no full nine-entry candidate through its root-box bound $R \le 10^6$; see the project's revalidation report. That external computation was source-checked here but not reproduced. Its root-box height is not interchangeable with any of the local bounds above.

Every scanner has synthetic positive tests and a small-range independent reference enumerator. The repository's regression suite currently contains 46 tests covering the finite classifications, arithmetic stages, Gaussian arithmetic, native scanner builds, and reference comparisons.

9 Reproducibility and proof map

From the repository root, run

```
make test
make paper
```

The first command executes the Python regression suite; several tests compile the C++20 scanners in temporary directories. The second command uses Tectonic to rebuild this report.

The principal proof files are:

File	Content
`block-balance.md`	local valuation spectrum, block inequalities, dominant exceptions, Legendre filters
`two-block-exponent-one.md`	arbitrary-exponent two-block exclusion and final 2-isogeny descent
`hourglass-sunit.md`	primitive denominators and frustration of 111/211 relations
`composite-blocks.md`	compatible orientation grouping and general common-factor bounds
`three-block-squarefree.md`	28 physical classes, arithmetic exclusions, and arbitrary-exponent transfer
`three-block-p2qr.md`	134-class weighted residue and the complete 118-stage filter proof
`literature-audit.md`	source status, overlap, and novelty cautions

This separation is deliberate. The report makes the scope and dependency graph readable; the companion notes preserve enough algebra to audit signs, units, valuations, exceptional cases, and elliptic-curve maps line by line.

10 Assessment and open review questions

The results appear materially stronger than a routine first computation. The two-block theorem allows arbitrary exponents; the three-block theorem transfers an exact physical classification and several descents beyond the squarefree case; and the p^2qr work exposes a finite, reproducible frontier. At the same time, no systematic MathSciNet or zbMATH priority search has been performed, and the methods overlap with established Gaussian-integer and elliptic-curve approaches. The appropriate public claim is therefore "independently checkable partial progress," not "new solution" and not a priority assertion.

The most useful independent reviews would address:

1. the unit and sign alignment in the coupled two-block edge relations;
2. the transfer from squarefree to arbitrary balanced/extreme blocks;
3. completeness of the physical symmetry quotient in the three-block classifiers;
4. the local-to-global step in each common-factor quotient;
5. the elliptic-curve descents and exceptional inverse images;
6. whether any of the 16 p^2qr survivors can be eliminated by a coupled identity or realized locally at all primes.

Acknowledgment

The public problem page at UnsolvedMath and the sources below provide context for independent discussion. Uwe Schwarz contributed human direction, computing resources, curation, and review without claiming mathematical authorship. The project actively invites independent verification and correction.

References

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